

EvoSeer

A biologically-informed stochastic evolutionary model

1) Introduction

1.a) Skin composition and structure

The skin is an organ composed of multiple layers and cell types. The outermost layer, the epidermis, is primarily made up of keratinocytes, which are responsible for the skin's protective barrier function. Melanocytes, which produce the pigment melanin, are also found in the epidermis and play a crucial role in protecting the skin from ultraviolet (UV) radiation. The dermis, located beneath the epidermis, contains fibroblasts, immune cells, blood vessels, and other structures that support the skin's function and integrity.

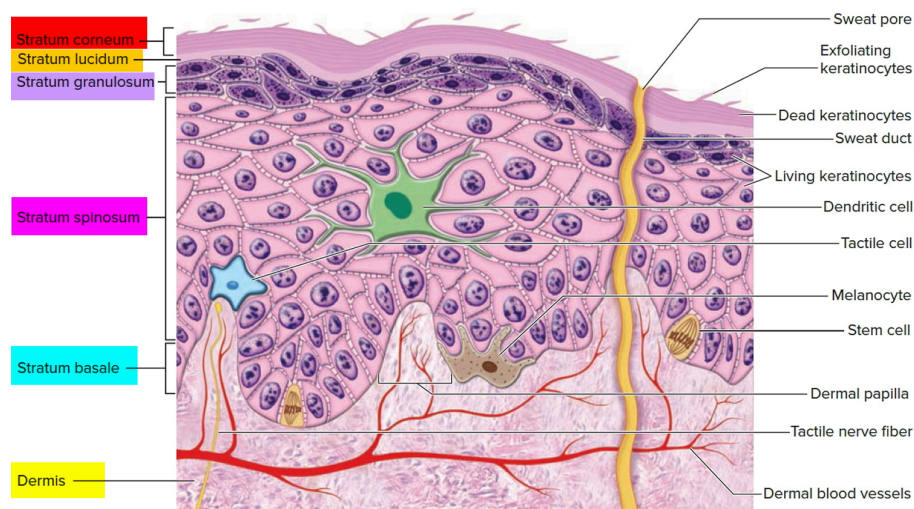


Figure 1. Structure of the epidermis. From random site.

Melanocytes are specialized, neural crest-derived cells responsible for the production of melanin, the pigment that determines skin and hair color. In the epidermis, melanocytes reside at the dermal-epidermal junction, where their numbers are maintained constant throughout life [1]. Each epidermal melanocyte extends dendrites that allow it to physically contact approximately 36 surrounding keratinocytes [1], forming a functional unit through which mature melanin-containing organelles called melanosomes are transferred to keratinocytes. Once internalized, these melanosomes are arranged by keratinocytes into parasol-shaped supranuclear caps that shield genomic DNA from further UV-induced damage [1]. Beyond the interfollicular epidermis, a reservoir of undifferentiated **Melanocyte Stem Cells (MSCs)** resides in the hair follicle bulge, where they remain quiescent and supply mature melanocytes to the follicle bulb during each hair cycle, as well as to the epidermis following UV- or wound-induced damage [1].

At the molecular level, melanocyte biology is governed by the RAS–ERK1/2 signaling pathway, a major driver of cell proliferation and survival. In normal melanocytes, this pathway is coordinately activated by two axes — the SCF–KIT pathway and the α -MSH–MC1R pathway — which both converge on the transcription factor MITF to balance survival, proliferation and melanin production [1]. This central role of the MAPK pathway in melanocyte biology is further underscored by the fact that the most common mutations in melanocytic tumors affect this pathway.

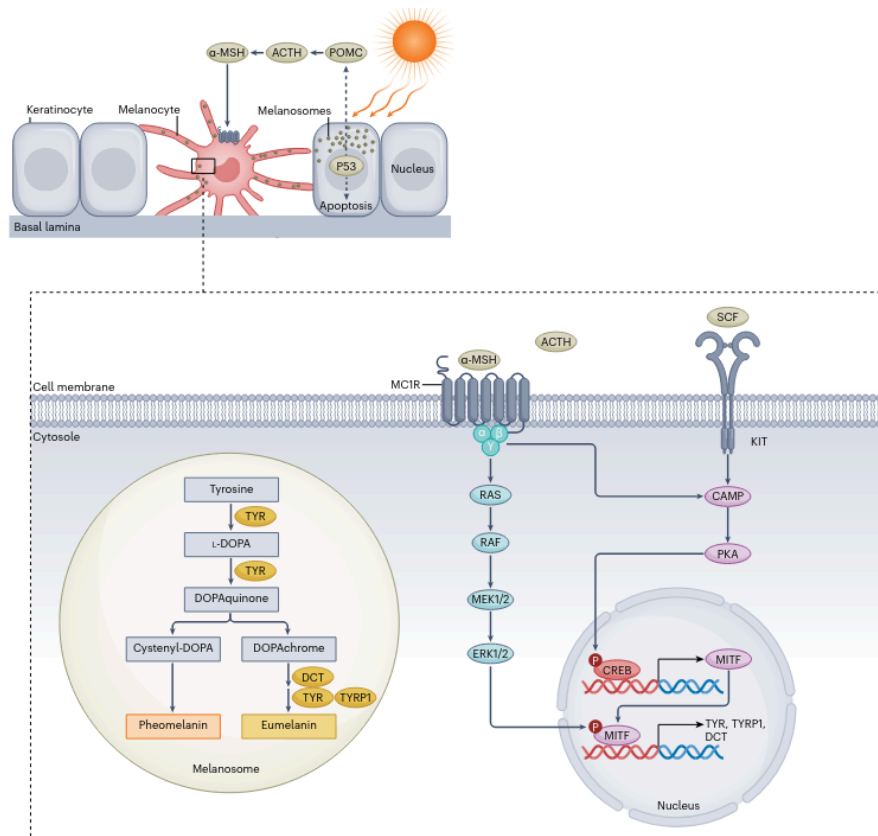


Figure 2. Molecular regulation of melanin production. Upon UV-induced DNA damage, keratinocytes activate TP53, triggering POMC processing into α -MSH, which signals through MC1R to elevate cAMP and activate CREB. In parallel, the SCF-KIT pathway converges on the same axis. Both pathways ultimately upregulate MITF, the master transcription factor driving expression of melanin-synthesis enzymes (TYR, TYRP1, DCT). Melanosomes carrying the resulting melanin are transferred via dendrites to keratinocytes, where they form supranuclear caps shielding DNA from further UV damage. Adapted from [1].

1.b) Melanocyte to melanoma

1.b.a) Nevi

Nevi, commonly known as moles, are benign proliferations of melanocytes. They can be congenital or acquired. While most nevi are harmless, some can serve as precursors to melanoma, a malignant tumor of melanocytes. Nevi initiation arises from the acquisition of mutations in the MAPK signaling pathway. Usually BRAF^{V600E} in 80% of nevi or NRAS activation is around 6-15% [2]. Nevi is regarded as one of the clearest example of cell senescence, as the acquisition of these mutations lead to a strong proliferation of melanocytes, which is then followed by a growth arrest. This growth arrest is due to the **Oncogen Induced Senescence (OIS)** mechanism, which is a cellular response to oncogenic stress that leads to a stable cell cycle arrest. OIS is thought to be a protective mechanism that prevents the uncontrolled proliferation of cells with oncogenic mutations, thereby acting as a barrier to tumorigenesis.

However all the cells present in the nevi are not necessarily senescent since some of them continue proliferating to compensate cell loss.

1.b.a.a) Oncogen Induced Senescence (OIS)

In human, there are two main intracellular molecular pathways effecting cell senescence in response to oncogenic stress by arresting the cell cycle. Respectively regulated by proteins p53 and p16 (Figure 3). Interestingly both proteins are encoded by the same gene CDKN2A by alternative splicing. This gene is mutated in around 20% of familial melanoma.

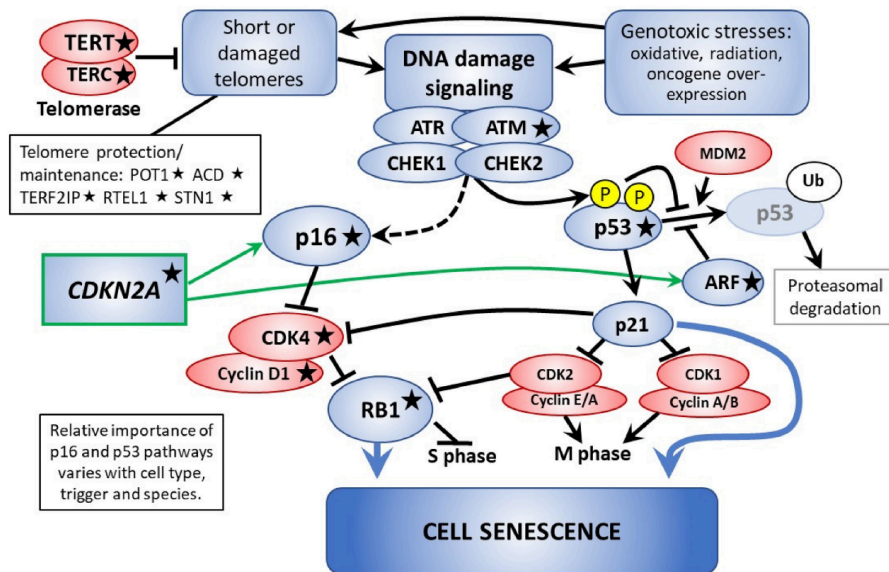


Figure 3. Main human cell senescence pathways, via p16 and p53. From [3]

1.b.b) Melanoma

Melanoma is a highly aggressive form of skin cancer that arises from melanocytes. It is characterized by its ability to metastasize rapidly and its resistance to conventional therapies. The progression from nevi to melanoma is a complex process that involves the accumulation of genetic and epigenetic alterations, as well as changes in the tumor microenvironment. Despite the role and protective function of OIS, some nevi can escape this growth arrest and progress to melanoma, particularly when additional mutations are acquired in genes that activate the OIS such as TERT promoter, CDKN2A, PTEN, TP53, etc.

1.c) Evolution models

In this state-of-the-art section, models are classified into two categories: those that rely on abstract representations of mutations, and those that incorporate real biological features of mutations. Note that we will only mention models post 2015 since a complete review has been written in 2015 by N. Beerenwinkel, R. F. Schwarz, M. Gerstung, and F. Markowetz [4]

TODO :

- <https://www.sciencedirect.com/science/article/pii/S2405471224003491>
- N. Beerenwinkel, R. F. Schwarz, M. Gerstung, and F. Markowetz [4]
- <https://link.springer.com/article/10.1007/s12032-023-01968-0>

1.c.a) Abstract evolution models

1.c.a.a) C. D. McFarland, L. A. Mirny, and K. S. Korolev [5]

This evolution model assigns each mutation an abstract fitness score rather than grounding it in specific biological properties. The majority of mutations carry a small negative fitness effect, while a minority confer a large positive fitness effect. This distribution mimics the observed imbalance between the few driver mutations that increase cellular fitness and the abundant passenger mutations that slightly decrease it.

1.c.a.b) A. Niida, T. Hasegawa, H. Innan, T. Shibata, K. Mimori, and S. Miyano [6]

Very similar to what we have already done but give inspiration of how to presents results. (A cell either die or divide and give driver/passenger generic mutation)

1.c.b) Biology Informed Models

1.c.b.a) J. Jiménez-Sánchez et al. [7]

3D, agent-based but agent is a subpopulation not a cell. Take in consideration croissanse dynamic, migration, mutation, death. But mutations are : activation or inactivation of a gene. Not individual effect of mutation - mutation probability dependant of Cosmic signature probability 3D, multi-compartmental and stochastic, aimed at reaching clinically realistic tumor sizes (100 cm³) at low computational cost. The agent is not an individual cell but a clonal subpopulation occupying a voxel, in the spirit of community-ecology models. Each subpopulation is defined by a binary trait vector

of length G (presence/absence of an alteration) and undergoes four stochastic processes — reproduction, death, migration, mutation — with synchronous updating and probabilities sampled from binomial distributions. Biological realism enters through weights w_i that modulate how each alteration shifts the four rates. Applied to GBM, the model uses $G = 3$ pathways (RTK/PI3K/RAS, RB, p53, from TCGA), yielding 8 possible genotypes; weights are calibrated by Approximate Bayesian computation (ABC) rejection against clinical mutation frequencies.

1.c.b.b) M. Ponce-de-Leon et al. [8]

PhysiBoSS is interesting because it's an Agent-Based Modeling (ABM) with several dedicated solvers: a diffusion and transport solver to simulate the chemical microenvironment and its interaction with the cells, a mechanical solver for cell–cell mechanical interactions, and other components to simulate cellular processes including growth, division, differentiation, and death. Each agent uses MaBoSS, a stochastic continuous-time simulator of Boolean intracellular networks. We should seriously investigate the possibility of integrating/mimicing this framework into our simulation as disgressed in Section 2.b.b.a . E.g., we could create our own model of melanoma and change gene basal and maximum activity based on the mutations each cell harbours.

1.d) Goal of the research

The goal of the research is to create a biologically informed stochastic evolutionary model. This model take as input the genomic landscape of a patient and simulate multiple melanocytes evolution. This “digital twin” of the patient will be used to predict the risk of melanoma acquisition based on the patient genomic landscape. This is particularly useful for patients with a high risk of melanoma, such as those with familial history of melanoma or those with a high number of nevi. The model will also be used to predict the risk of melanoma progression and to identify potential therapeutic targets.

Moreover this model can be more generally used to predict the risk of acquisition and progression of other types of cancer, such as lung cancer, breast cancer, etc. Following the training of the model we are able to predict the features importance for each tumor type, which can be used to identify potential therapeutic targets and to develop personalized treatment strategies.

Moreover the model allow us to predict the order of acquisition of mutations.

2) Method

2.a) Model description

The model describe a population of cells that evolve through time. It is a stochastic model in which each cell is characterised by the set of mutations it harbours (Figure 4). Theses mutations impact its biological parameters (proliferation, immunogenicity, mutation rate, etc.) that will impacts their death-birth parameters.

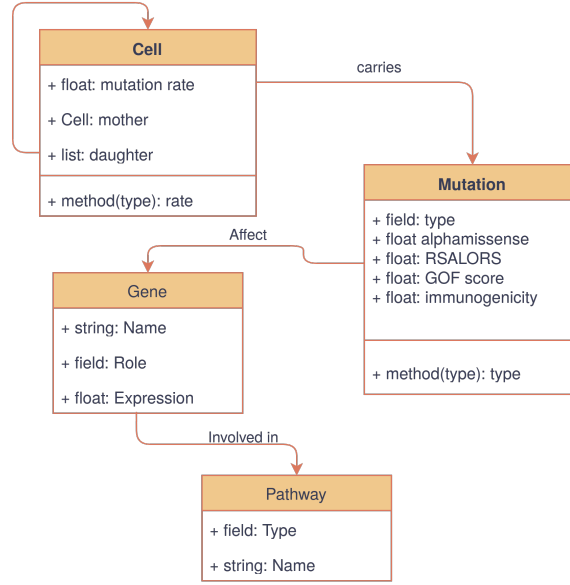


Figure 4. Schematic representation of the model. Each cell carries mutations that impact its biological parameters depending of the gene the mutation impact and the type of mutation. Relations between mutations and parameters are described in the following section.

When a cell divides, it gives birth to a daughter cell that inherits the same set of mutations as the parent cell, plus a number of de-novo mutations that are acquired during the division. The parameters of the daughter cell are then updated based on the effect of the inherited and de-novo mutations. Each cell can also die, with a probability that depends on its parameters and the population size. The model is simulated through time, allowing us to track the evolution of the cell population and to predict the risk of melanoma acquisition and progression based on the patient genomic landscape.

2.a.a) Mutation rate

The number of de novo mutations acquired by a daughter cell depends on the mutation rate μ_i of its parent, which reflects the cell's capacity to repair DNA damage. We model μ_i as a function of **Loss-of-Function (LOF)** mutations accumulated in genes belonging to **DNA repair mechanism** pathways, namely:

- Nucleotide Excision Repair (NER)
- Base Excision Repair (BER)
- Mismatch Repair (MMR)
- Homologous Recombination (HR)
- Non-Homologous End Joining (NHEJ)
- Direct Repair (DR)
- Microhomology-Mediated End Joining (MMEJ)
- Single-Strand Annealing (SSA)

Since driver mutations in **DNA repair mechanism** are by nature **LOF**, μ_i is defined as the number of pathways harboring at least one **LOF** mutation:

$$\mu_i = \sum_{p \in \mathcal{P}} \mathbb{1}[\text{LOF}_{i,p}] \quad 2.1$$

where $\mathcal{P} = \{\text{NER, BER, MMR, HR, NHEJ, DR, MMEJ, SSA}\}$ denote the set of **DNA repair mechanism** pathways considered. (This list is exhaustive with respect to N. Chatterjee and G. C. Walker [9] and will likely be simplified during implementation.) $\mathbb{1}[\text{LOF}_{i,p}]$ is the indicator that at least one gene of pathway p carries a **LOF** mutation in cell i .

2.a.b) Event rate

The probability that an event (death or division) occurs for a cell i is computed as the rate of event of the cell λ_i divided by the sum of the rate of event of all cells in the population, following Equation 2.2.

$$P(\text{event for } i) = \frac{\lambda_i}{\sum_{j=1}^N \lambda_j} \quad 2.2$$

$$\lambda_i = \sum_{f=1}^F w_f \cdot S_{f,i} \cdot N^{\alpha_f}, \quad \alpha_f \in \mathbb{R} \geq 0 \quad 2.3$$

Where w_f is the weight of the feature f , S_f is the value of the feature f for the cell i , and N^{α_f} is a term that accounts for the population size effect on the rate of event. α_f and w_f being the parameters of the model that will be learned during the training phase.

Each of the features f is categorized as either a “proliferative” feature ($f \in \mathcal{B}$), which increases the birth rate of the cell, or a “deleterious” feature ($f \in \mathcal{D}$), which increases the death rate of the cell. A summary table of these features is available at Table 1.

$$b_i = \sum_{f \in \mathcal{B}} w_f \cdot S_{f,i} \cdot N^{\alpha_f}, \quad d_i = \sum_{f \in \mathcal{D}} w_f \cdot S_{f,i} \cdot N^{\alpha_f} \quad 2.4$$

Based on this, when a cell is selected to undergo an event based on Equation 2.2, the cell either divides with probability $P(\text{divide} | i) = \frac{b_i}{\lambda_i}$ or dies otherwise.

2.a.c) Timing

Intuitively, the total rate of event $\Lambda = \sum_{j=1}^N \lambda_j$ can be view as the number of events that occur in the population per unit of time. Therefore, we can compute the time between two events as a function of the rate Λ .

In the standard Gillespie algorithm, the time increment between two consecutive events is not constant but is drawn from an exponential distribution parameterized by Λ :

$$\Delta t = -\frac{\ln(U)}{\Lambda}, \quad U \sim \text{Uniform}(0, 1) \quad 2.5$$

By accumulating Δt at each iteration, one obtains an absolute simulation time $t \leftarrow t + \Delta t$ that maps to real biological time.

2.b) Cell parameters

As briefly described in Figure 4, each mutation is characterized by its effect on a set of features that impact the cell's parameters. For example, a mutation in the gene BRAF may increase the proliferation rate of the cell, while a mutation in the gene TP53 may decrease the death rate of the cell. We describe below the cell parameters and how mutations modulate them depending on the gene and pathway they affect. Note that these descriptions and mathematical implementation are ideas and first research tracks that are subject to modification and ajustement as the research will move forward.

Proliferative ($f \in \mathcal{B}$)	Deleterious ($f \in \mathcal{D}$)
MAPK Pathway (P_{MAPK} Equation 2.8)	Immunogenicity (Φ Equation 2.7)
PI3K Pathway (P_{MAPK} Equation 2.8)	Apoptotic Pathway (P_{apop} Equation 2.8)
OIS Pathway (function of λ_i^{OIS} Equation 2.10)*	

Table 1: Summary of the proliferative and deleterious features f used in the model. * OIS pathway activation has a negative effect on the proliferation but no effect on the death.

2.b.a) Immunogenicity

Each no-synonymous mutation create new neo-antigen that can potentially be recognized by the immune system. This explain why the high-TMB tumors are more immunogenic and more responsive to immunotherapy than low-TMB tumors since it harbors more neo-antigens and are more likely to be recognized by the immune system. However, not all neo-antigens are equally immunogenic. Following the formula from M. Łuksza *et al.* [10] , we compute the immunogenicity of a cell as the product of the amplitude and recognizability of its neo-antigens Equation 2.6.

$$\varphi = \alpha \cdot R \quad 2.6$$

$$\alpha = \frac{1/\text{IC50}_{\text{mt}}}{1/\text{IC50}_{\text{mt}} + 1/\text{IC50}_{\text{wt}}} \quad R = \max_{\text{epitope}} \exp(-k \times d(\text{peptide}, \text{epitope}))$$

Where the amplitude α measure how much stronger the mutant peptide binds MHC-I compare to the wild-type peptide, and the recognizability R measure the similarity of the mutant peptide to known pathogen-derived T-cell epitopes from the IEDB database [11] as scored by a BLOSUM substitution matrix.

Note 2.2.1.1 (Preliminary Observation).

In agreement with D. Hoyos *et al.* [12], we observe a trade off between immunogenicity and pathogenicity. Indeed we use the alphasense score as a proxy of the pathogenicity of the mutation, and we observe a small spearman correlation of 0.25 between the alphasense score and φ .

Note 1: Preliminary Observation

On the other hand, the cell can accumulate mutation that allow it to escape the immune system, such as HLA loss, JAK mutations, PD-L1 upregulation, etc. We model this immunosuppression as a parameter that controbalance the immunogenicity of the cell. Overall the immunogenicity of the cell Φ is the sum of the immunogenicity of its neo-antigens contrebanced by its immunosuppression level, which reflect the presence of immune escape mechanisms such as HLA loss, JAK mutations, PD-L1 upregulation, etc.

$$\Phi = \sum_{m=1}^M (\varphi_m) \cdot 1 - immunosuppression \quad 2.7$$

Where the immunosuppression level has to be defined.

2.b.b) Pathways

In order to take epistatic interactions between mutations into account, we model the proliferation of the cell as a function of the activation state of several signaling pathways relevant to melanoma progression: the MAPK pathway, the PI3K pathway and the apoptotic pathway [13]. Each pathway is activated by a set of genes that can harbor **Gain-of-Function (GOF)** or **LOF** mutations, and the activation level of pathway k is computed as a sigmoid function of the weighted contribution of its constituent mutations, following Equation 2.8.

$$P_k = \sigma_k \left(\sum_{g \in \text{act}_k} \omega_g^k \cdot x_g^{\text{GOF}} - \sum_{g \in \text{inh}_k} \omega_g^k \cdot x_g^{\text{LOF}} \right) \quad 2.8$$

Where $x_g^{\text{GOF}} \in \{0, 1\}$ indicates whether the cell carries an activating mutation in gene g (resp. x_g^{LOF} for inactivating mutations), ω_g^k is the contribution weight of gene g within pathway k , and σ_k is a sigmoid function with pathway-specific slope and threshold. The sigmoid ensures that saturation within a pathway produces diminishing returns: a cell already harboring BRAF^{V600E} gains limited additional MAPK activation from subsequent NF1 loss, recovering the mutual exclusivity observed in tumor genomic data [14], [15].

2.b.b.a) Pathway structure and weighting

We represent each pathway as a weighted tree, where leaf nodes correspond to individual genes and internal nodes correspond to sub-pathways or signaling hubs (e.g., RAS as an integration node upstream of RAF-MEK-ERK). The weight of a node represents its fractional contribution to its parent node, with weights summing to 1 at each level within a pathway. This normalization makes weights interpretable as relative contributions rather than arbitrary coefficients.

Three strategies are considered for setting the leaf weights ω_g^k :

1. **Uniform leaf weights.** All genes at the leaves receive equal weight within their sub-pathway. Because of the tree structure, hub genes that integrate multiple downstream effects mechanically acquire larger effective weights than peripheral genes. This is the most parsimonious option and avoids gene-by-gene parameter inflation.
2. **Tiered weights.** Genes are assigned to three categories based on their role in the pathway: high weight (e.g., $\omega = 0.5$) for dominant drivers in the melanocyte context (BRAF in MAPK, CDKN2A in cell cycle), medium weight (e.g., $\omega = 0.3$) for significant but partially redundant contributors (NRAS in MAPK), and low weight (e.g., $\omega = 0.1$) for modulators and accessory regulators (RASA2, SPRED). Weights are then normalized within each sub-pathway. This reduces the effective number of free parameters to three tier values, which can be either fixed from the literature or treated as ABC parameters.
3. **Literature-derived weights.** Each ω_g^k is fixed from an external source. Candidate sources include gene essentiality data in melanoma cell lines (DepMap/Project Achilles), pathway-specific curated databases (Reactome, SIGNOR), or tissue-specific expression (GTEx) — though each proxy has known limitations, in particular for tumor suppressors whose function is regulated post-translationally.
4. **Markovian Boolean Stochastic Simulator:** Instead of using approach from Equation 2.8 we will use the approach of M. Ponce-de-Leon *et al.* [8] in PhysiBoSS by using a Markovian Boolean Stochastic Simulator (such as MaBoSS) that give a probability for a pathway to be activated depending of the basal and maximum activity of gene in this pathway. We can imagine giving a score of basal activity and maximum activity of genes based on the mutation it harbors.

The choice of strategy will depend on the identifiability of the weights during ABC calibration and on the sensitivity of the simulation outputs to individual ω_g^k values.

2.b.c) Oncogen Induced Senescence (OIS) pathway

The OIS pathway is modeled as a balance between two decoupled components: a **genotypic susceptibility** (S_i) (Equation 2.9) encoding whether a cell has a functional OIS machinery, and a **progression variable** (Section 2.b.c.b) encoding how close the cell is to triggering OIS given its history. The fire event is stochastic, with an instantaneous hazard given by the product of these two components. This decoupling is motivated by the biological observation that OIS fails in two distinct ways: (i) genetic disruption of the OIS effector (e.g., biallelic CDKN2A loss [16]), which prevents OIS from ever being triggered; and (ii) absence of oncogenic drive (wild-type melanocytes), which leaves the OIS machinery intact but unengaged.

2.b.c.-a) Genotypic susceptibility

Following the formalism of the other pathways (Equation 2.8), the OIS susceptibility of a cell i is computed as a sigmoid of the weighted contribution of pro- and anti-senescence mutations.

$$S_i = \sigma \left(\sum_{g \in \text{OIS}_{\text{act}}} \omega_g \cdot x_g^{\text{GOF}} - \sum_{g \in \text{OIS}_{\text{inh}}} \omega_g \cdot x_g^{\text{LOF}} \right) \in [0, 1] \quad 2.9$$

Where the sets OIS_{act} (pro-senescence genes) and OIS_{inh} (anti-senescence genes) are curated from the CellAge database [17], filtered to retain genes with documented involvement in melanocyte senescence.

More intuitively we can see a WT cell with no oncogeneic activation has a baseline $S_i \approx \sigma(0) = 0.5$ (OIS machinery armed but unengaged), a $\text{BRAF}^{\text{V600E}}$ cell has strong positive terms and $S_i \rightarrow 1$ (fully armed), a $\text{BRAF}^{\text{V600E}} + \text{CDKN2A}^{-/-}$ cell has a broken anti-senescence mechanism and $S_i \rightarrow 0$ (OIS cannot fire). The logic of OIS escape by LOF in effector emerges naturally from this formulation rather than requiring a separate mechanism.

2.b.c.a) stochastic firing

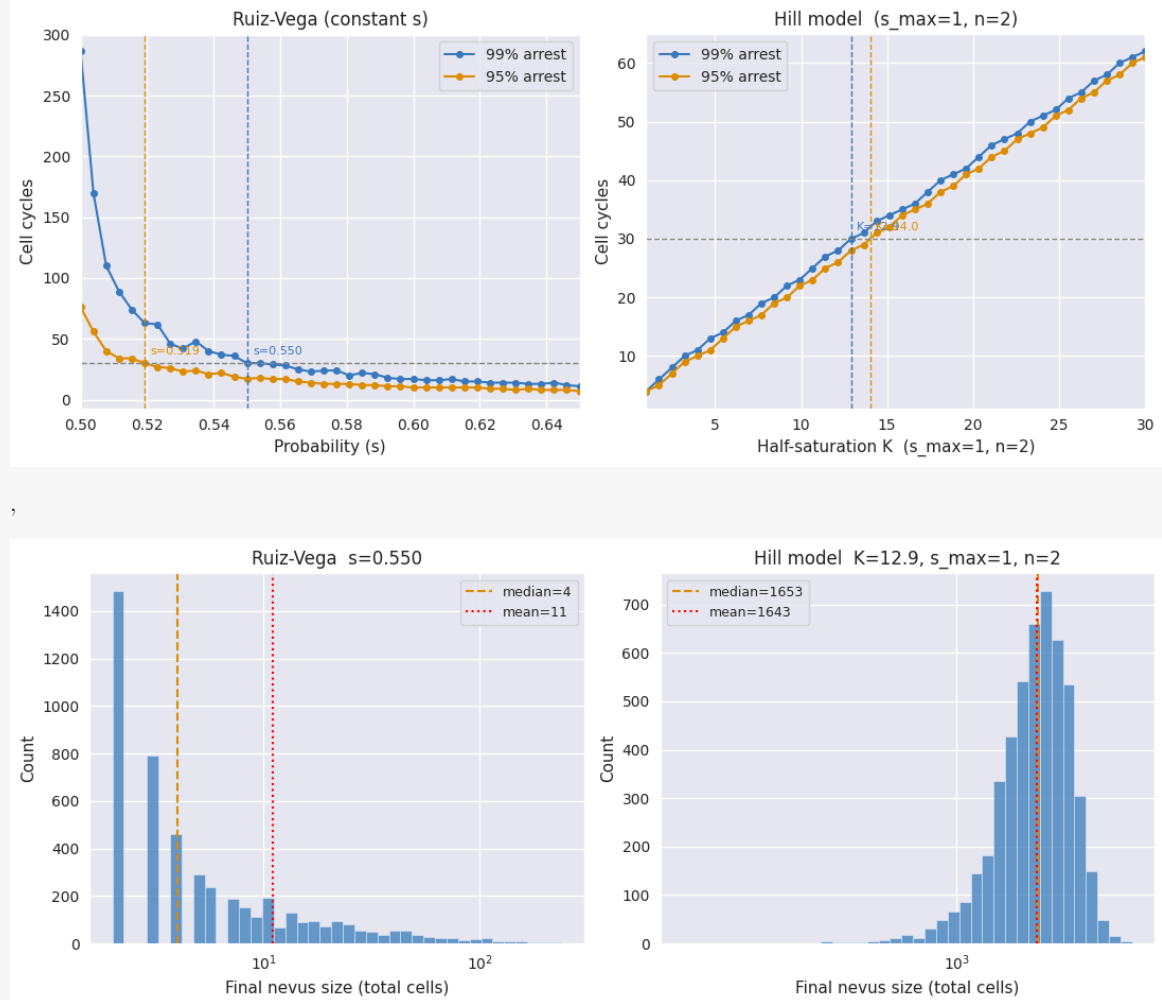
The OIS trigger event fires with a hazard (rate per unit time) that combines the **genotypic susceptibility** and a progression-dependent multiplier.

$$\lambda_i^{\text{OIS}}(t) = S_i \cdot \lambda_0 \cdot \text{Hill}(u_i(t), K, n) \quad 2.10$$

where $u_{i(t)}$ is the progression variable of cell i , K is the half-saturation point, n is the Hill coefficient, and λ_0 sets the overall time scale. The Hill form is motivated by preliminary simulations (see Note 2) showing that a constant per-division arrest probability [18] produces an exponential clone size distribution, whereas a Hill-saturating probability produces a log-normal-like distribution consistent with empirical nevus size distributions.

Note 2.2.3.1.1 (Preliminary Observation).

I quickly tested this implementation ([code here](#)). First I reimplemented what they did and fall back on the same results and after I implemented the Hill function and modify K. Notably, this modification yields an more like log-normal size distribution for the Hill implementation. Playing on the n and $s_{\max}$ should allows us to further improve the model.



Note 2: Preliminary Observation

2.b.c.b) Progression

Two hypotheses for the progression variable u_i are formulated and will be compared.

2.b.c.b.a) H1 - Time-driven progression.

$u_{i(t)} = t - t_i^{\text{trigger}}$, where t_i^{trigger} is the time at which cell i ' lineage acquired its first OIS-triggering mutation. Under this hypothesis, the accumulation of senescence-inducing stress is assumed to occur on a wall-clock timescale, reflecting processes such as **Senescence-Associated Secretory Phenotype (SASP)** buildup, oxidative damage, or metabolic stress that progress independently of replication.

2.b.c.b.b) H2 - Division-driven progression with oncogenic coupling.

$u_{i(t)} = k_{i(t)}$, the number of divisions undergone by the lineage of cell i since the acquisition of its first OIS-triggering mutation. The half-saturation K is itself modulated by the oncogenic signal: $K(P_{\text{MAPK}}) = \frac{K_0}{P_{\text{MAPK}}^{\gamma}}$, so that stronger oncogenic activation accelerates the approach to arrest. under this hypothesis, the driving force is replicative stress (stalled forks, DNA damage response activation, telomere dysfunction), which accumulates through division cycles rather than real time. The two hypotheses are not mutually exclusive under all conditions: if division rate is approximately constant, k_i and τ_i are proportional and H1/H2 become empirically indistinguishable. They diverge when the division rate varies substantially — notably in the early hyper-proliferative phase following driver acquisition, where H2 predicts faster progression than H1.

2.b.c.c) Division counter propagation

At each division of cell i into daughters i and i' the counter is propagated as: $k_i = k_{i'} = k_i + 1$. Both daughters inherit the incremental counter, reflecting the fact that each chromosomal copy has undergone the same number of replication events and accumulated comparable replicative stress. The counter resets to 0 only upon acquisition of a new **OIS**-triggering mutation that dominates the signal. The phenotypic stability of mature nevi (constant size over years to decades) cannot be produced by a population entirely in **OIS** arrest: some cells must continue to divide to compensate for cell loss. We model this as a structurally resistant subpopulation of proliferative-reserve cells (working hypothesis, to be challenged — cf. the “regenerating nevus” concept of R. E. Vilain, S. W. McCarthy, and R. A. Scolyer [19], and the evidence from X. Wang *et al.* [20] that senescent melanocytes influence neighboring stem-cell-like compartments).

Operationally, a small fraction $p \ll 1$ of cells arising from division are flagged as proliferative-reserve. These cells have their **OIS** hazard suppressed ($S_i \rightarrow 0$ regardless of genotype) and divide at a reduced baseline rate. They are the main source of mutation accumulation once the bulk of the clone has arrested, and therefore the main substrate for acquiring escape mutations that drive transformation. The alternative hypothesis — that stability emerges from stochastic escape from a homogeneous **OIS** hazard — is also testable and will be compared.

2.b.c.d) Oncogen Induced Senescence (OIS) escape and transformation

Escape from the arrested state is not modeled as a separate event but emerges from the same formalism: a proliferative-reserve cell whose lineage acquires sufficient LOF mutations in **OIS** effectors (primarily CDKN2A, TP53) will see its S_i drop below the threshold required for hazard firing. Its descendants therefore never arrest, regardless of their accumulated division count, producing a new wave of clonal expansion from within the stable nevus. This mechanism corresponds to the mechanistic criterion for the melanoma terminal regime (Section 2.c.b).

2.c) Training and validation

2.c.a) General strategy

The model is calibrated and validated in two stages, each targeting distinct structural components :

1. **ABC calibration**: The global parameters - feature weights w_i and exponents α_f (Equation 2.3) and mutation rate μ_i are calibrated via **ABC** against summary statistics extracted from the ISIC 2024 dataset [21]. (Section 2.c.c)
2. **External validation** : The calibrated model is evaluated on three independent targets, each stressing a different structural component of the model (Section 2.c.d)

2.c.b) Simulation outcomes and phenotypic classification

Each simulation produces a population trajectory $N(t)$ starting from one or a few wild-type melanocytes. At the end of the simulation horizon, each trajectory is classified into one of three categories :

- **No tumor**: Extinction or no sustained expansion beyond stochastic fluctuations.
- **Benign nevus**: Limited expansion reaching a stable plateau.
- **Melanoma**: Sustained expansion without plateau within the simulation window.

2.c.c) ABC calibration target

The primary summary statistic for **ABC** calibration is the size distribution of benign nevi from ISIC 2024 [21], under the proxy projected lesion area $\approx$ number of cells. A complementary summary statistic is the relative proportion of the three terminal regimes across independent simulation runs, which carries information on the global balance between **OIS**, immune pressure and proliferative drive, and helps break the known degeneracy between **OIS** kinetics and immunogenicity in the stable nevus regime. We acknowledge and assume the following biases of this target:

- **Acquisition bias**. ISIC 2024 (SLICE-3D) is derived from dermatological consultations, enriching for high-nevus-count and light-phototype individuals. The distribution is not representative of the general population.
- **Left-truncation**. Clinically undetectable nevi are absent from the dataset. The calibration therefore fits a conditional distribution given detectability.
- **Age heterogeneity**. ISIC aggregates nevi across patients and ages. We assume that the expansion window of a nevus is short relative to typical observation ages, so that observed nevi have reached their mature plateau and can be compared to the simulator output at horizon T .

2.c.d) External validation

Three out-of-sample targets are used to validate the calibrated model, each probing a distinct structural component:

1. **CDKN2A germline familial melanoma — validates the **OIS** module**. Patients carrying germline loss-of-function mutations in CDKN2A (p16/p14ARF) exhibit early-onset, multiple primary melanomas with well-documented incidence and age-of-onset distributions (e.g., GenoMEL cohorts [22]). The **OIS** parameters of our model are calibrated on the general-population nevus regime, without exposure to CDKN2A-deficient input. Simulating populations with

a CDKN2A-deficient germline background and comparing the predicted melanoma incidence and onset distribution to published cohorts therefore constitutes a genuine out-of-sample test of the **OIS** module.

2. **Xeroderma Pigmentosum (XP)** — **validates the mutation-rate module** **XP** patients carry germline defects in **NER**, resulting in a dramatically elevated effective mutation rate under UV exposure and an exacerbated UV mutational signature. **XP** cohorts show a median age at first melanoma of approximately 22 years versus 60+ in the general population, with an estimated 2000-fold increased lifetime prevalence [23] TODO ADD CITATION for age in the general population. Simulating populations with an elevated mutation rate parameter consistent with **NER** deficiency, and recovering the observed age-of-onset shift and incidence increase, validates the structure linking mutation accumulation to driver acquisition and transformation.
3. **Order of driver mutation acquisition** — **validates the transformation trajectory**. The phylogenetically reconstructed order of driver acquisition in established melanomas [14], [24] is compared to the order observed in the melanoma subset of our simulations. This validation is restricted to the transformation-positive regime, consistent with the fact that M. Gerstung *et al.* [24] reconstruction is based on clinically detected tumors and therefore reflects survival-conditioned sampling.

The three validation targets are partially coupled in their predictive requirements: in particular, **XP** predictions depend both on the mutation-rate module and on the **OIS** module (since accumulated mutations must produce second hits in **OIS** effectors to drive transformation). Clean diagnostic separation is therefore imperfect, but the three targets jointly probe a sufficiently broad region of model behavior to constitute meaningful out-of-sample tests. Je pense mettre xp dans training ou alors order of driver + CDKN2A en training et valider sur XP puisque il prends les deux en compte

2.c.e) Model parsimony

The number of free **ABC** parameters is deliberately kept small. This choice is motivated by the identifiability constraints of the chosen summary statistics: the ISIC size distribution and the terminal-regime proportions jointly constrain only a limited number of effective dimensions in parameter space. Introducing additional free parameters beyond this identifiable subspace would yield posterior distributions dominated by the prior, without improving predictive performance. Parameters that cannot be identified from the current summary statistics are fixed a priori rather than included in the inference.

3) Work plan

3.a) Nevi Evolution Model

As briefly show in the Gantt Chart of the project (Figure 5), we will first focus on the nevi evolution model since its implementation is lighter than the melanoma. E.g immunogenicity has negligible role in common melanocytic nevi [25]. This is also a first preliminary step to modelise melanoma. In this implementation we will mainly focus on the **OIS** pathway (Section 2.b.c) aswell as testing different pathways implementation (Section 2.b.b).

In addition this model have to put the basic of the EvoSeer framework. This model will be a crash-test to think about the best structure of the framework in order to easily add other features to predict cell fates aswell as mutation effect. At the end of the implementation I believe we will have interesting results to publish either on the clinical aspect (nevi expansion dynamic, difference of $N(t)$ trajectory depending of the first driver, ...) or/and on the framework aspect.

3.b) Melanoma Evolution Model

Building upon the nevi framework, the melanoma evolution model extends EvoSeer to capture the full trajectory from **OIS** escape to invasive disease. This requires integrating additional biological complexity that was deliberately excluded from the nevi model. Such as no-coding mutation (e.g. TERT promotor), immunity escape, ...

4) Work plan

4.a) Nevi Evolution Model

As briefly shown in the Gantt chart of the project (Figure 5), we will first focus on the nevi evolution model since its implementation is lighter than that of melanoma. For instance, immunogenicity plays a negligible role in common melanocytic nevi [25]. This also serves as a preliminary step toward modelling melanoma. In this implementation we will mainly focus on the **OIS** pathway (Section 2.b.c) as well as testing different pathway implementations (Section 2.b.b). In addition, this model has to lay the foundations of the EvoSeer framework. It will serve as a testbed to determine the best architecture for the framework, so that additional features for predicting cell fates and mutation effects can be incorporated later. At the end of this implementation, I believe we will have interesting results to publish, either on the clinical aspect (nevi expansion dynamics, differences in $N(t)$ trajectories depending on the initiating driver, ...) or on the framework aspect.

4.b) Melanoma Evolution Model

Building upon the nevi framework, the melanoma evolution model extends EvoSeer to capture the full trajectory from **OIS** escape to invasive disease. This requires integrating additional biological complexity that was deliberately excluded from the nevi model, such as non-coding mutations (e.g. TERT promoter), immune escape, and the multidimensional fitness vector coupling proliferative and deleterious features. At the end of this phase, EvoSeer should be able to simulate complete melanoma evolutionary trajectories — from the initial $\text{BRAF}^{\text{V600E}}$ event through **OIS**, escape, clonal diversification, and malignant progression — with each step parameterized by biologically realistic, mutation-specific features rather than generic rate constants.

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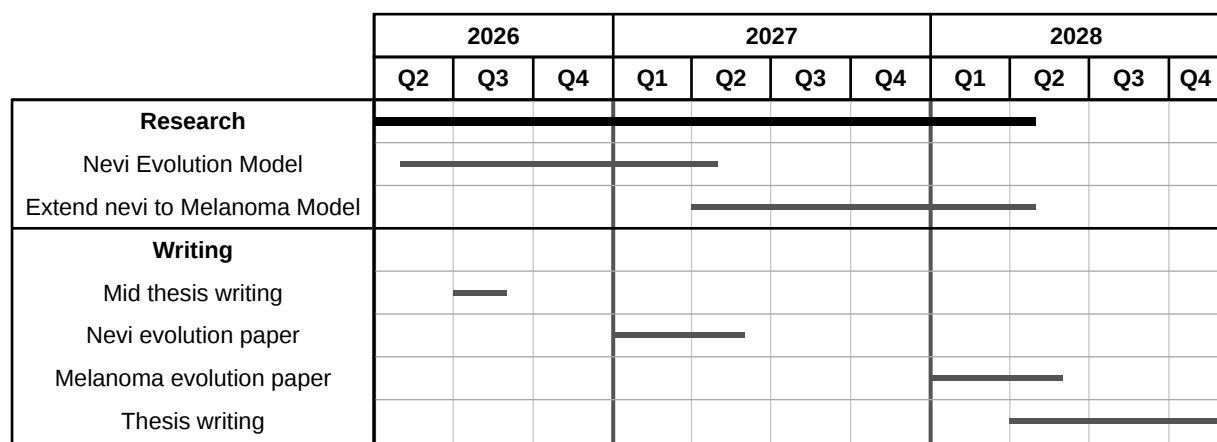


Figure 5. Thesis Work plan

Glossary

ABC – Approximate Bayesian computation: Constitutes a class of computational methods rooted in Bayesian statistic that can be used to estimate the posterior distributions of model parameters

ABM – Agent-Based Modeling: Modeling approach in which individual entities (agents) are simulated autonomously, each following defined rules and interacting with their environment and other agents, allowing emergent collective behaviors to arise from local interactions.

BER – Base Excision Repair: base excision repair pathway (BER)

DR – Direct Repair: direct reversal of DNA damage (DR)

DRM – DNA repair mechanism: mechanism that repair damages in the DNA

GOF – Gain-of-Function: Mutation that give a new or enhanced activity to the protein.

HR – Homologous Recombination: homologous recombination repair pathway (HR)

LOF – Loss-of-Function: Mutation that reduces or abolishes the normal activity of the protein.

MMEJ – Microhomology-Mediated End Joining: microhomology-mediated end joining repair pathway (MMEJ)

MMR – Mismatch Repair: mismatch repair pathway (MMR)

MSC – Melanocyte Stem Cells: Stem cells that give rise to mature melanocytes.

NER – Nucleotide Excision Repair: nucleotide excision repair pathway (NER)

NHEJ – Non-Homologous End Joining: non-homologous end joining repair pathway (NHEJ)

OIS – Oncogen Induced Senescence: A cellular response to oncogenic stress that leads to a stable cell cycle arrest.

SASP – Senescence-Associated Secretory Phenotype: The complex mixture of cytokines, chemokines, growth factors, and proteases secreted by senescent cells into their microenvironment, contributing to both immune-mediated clearance of senescent cells and chronic tissue inflammation.

SSA – Single-Strand Annealing: single-strand annealing repair pathway (SSA)

XP – Xeroderma Pigmentosum: Xeroderma pigmentosum (XP) is a rare inherited multisystem disorder characterized by a heightened sensitivity to the DNA damaging effects of ultraviolet radiation (UV) due to a default of nucleotide excision repair pathway (NER)

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